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SHOT TIMER

Abstract

Systems and methods for timing the discharge of a firearm using a shot timer are described herein. The shot timer uses multiple levels of verification. A first level includes analyzing the audio signal to determine if a sound received is associated with the discharging of a firearm. A second level includes analyzing the posture of individuals to eliminate individuals that likely are not the individuals that discharged the firearm. A third level includes analyzing the movement of one or more firearms to determine which firearm was the source of the audio signal.

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Background/Summary

BACKGROUND

[0001] Shot detection and timing systems can be used in various environments. For examples, shot

detection and timing systems can be used in shooting competitions. In these competitions, an audio, visual, or tactile start signal is used to start the competition. The shot timer is used to track a competitor's time by detecting the sound of shots fired by a weapon used by the competitors. The elapsed time between the start signal and the detected one or more shots is recorded and used to determine a competitor's score. A shot timer can also be used to train first responders or military personnel. For example, a shot timer can be used to track data such as how quickly trainees are performing certain actions such as unholstering a gun, shooting one or more targets, and reloading. [0002] Some conventional shot timers use one or more microphones to detect the sound of a shot being fired and record the elapsed time from the start signal to the detection of the sound. The shot timers use audio processing technologies that recognize the compressive sound of a shot being fired from a weapon. Some conventional shot timers are wearable devices that may use a combination of microphones and accelerometers to detect and record shooting data. The accelerometers can be used to sense movements such as a weapon being unholstered and holstered. However, these conventional systems can have some drawbacks. For example, in some situations, such as during training and competition, there may be more than one person firing their weapon during the same proximate time. If the weapons are fired simultaneously or nearly simultaneously with different calibers of weapons, conventional shot timers may not be able to identify which shooter fired which shot. Further, echoes or other loud noise may be detected as false positives. Additionally, in some situations, a shot timer may not detect that more than one shot was fired. Further, shot timers worn by a user can be difficult to synchronize, resulting in potentially erroneous data.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0003] FIG. 1 is an illustration of a range that uses a shot timer to detect and record firearm discharge times, in accordance with one or more examples of the present disclosure.

[0004] FIG. 2 illustrates an audio signal representing audio generating by a firearm and an audio signal generated by a person yelling near a shot timer, in accordance with one or more examples of the presently disclosed subject matter.

[0005] FIG. 3 illustrates an audio signal representing audio generating by a firearm and an audio signal stored in the sound data store, in accordance with one or more examples of the presently disclosed subject matter.

[0006] FIG. 4 is an illustration of a shot sound user interface configured to receive inputs to identify a sound signal for storage in a sound data store, in accordance with one or more examples of the presently disclosed subject matter.

[0007] FIG. 5 illustrates the use of skeleton tracking to determine which, if any, individual discharged a firearm, in accordance with one or more examples of the presently disclosed subject matter.

[0008] FIG. 6 illustrates a firearm in a pre-discharge position and a post-discharge position, in accordance with one or more examples of the presently disclosed subject matter.

[0009] FIG. 7 illustrates a method for timing discharges of a firearm, in accordance with various examples of the presently disclosed subject matter.

[0010] FIG. 8 depicts a component level view of the shot timer 102 for use with the systems and methods described herein, in accordance with various examples of the presently disclosed subject matter.

DETAILED DESCRIPTION

[0011] The presently disclosed subject matter is directed to a hybrid shot timer that uses multilayer verification technologies to verify shots (or discharges of a firearm). Each of the layers of verification are used to reduce the probability of having a false positive and, in some examples, to

identify which individual discharged their firearm, along with other advantages discussed below. In a first layer of verification, the hybrid shot timer uses audio processing technologies to discriminate between the sound a discharging firearm makes and other loud sounds. In a second layer of verification, the hybrid shot timer uses pre-discharge visual processing technologies to discriminate between individuals having body movements indicative of a person ready or currently discharging their firearm and other individuals either at rest or not discharging their firearms. In some examples, a camera of the hybrid shot timer uses skeletal tracking to identify and track those body movements as well as movements of the firearm. In a third layer of verification, the hybrid shot timer uses post-discharge visual processing technologies to discriminate between individuals in a ready position (i.e., ready to discharge their firearm) and the individuals that have discharged their firearm. In some examples, the post-discharge visual processing technologies use movements of the individual (e.g., such as someone experiencing a recoil of the firearm), movements of the firearm (e.g., the movement of the firearm's slide or the movement upwards of the firearm itself), and/or a muzzle flash to identify the individual associated with the sound of a firearm being discharged received at the hybrid shot timer.

[0012] Turning to the figures, whereby wherever possible the same reference numbers will be used throughout the drawings to refer to the same or like parts, FIG. 1 is an illustration of a range **100** that uses a shot timer **102** to detect and record firearm discharge times, in accordance with one or more examples of the present disclosure. In some examples, one or more of the functions of the shot timer **102** may be performed on a user device, a central server, or both the user device and the central server. In some examples, the shot timer **102** is one or more devices having computer-executable instructions stored thereon that perform the functions described herein. In further examples, the shot timer **102** may be an application on a user device (such as a cell phone) capable of performing the functions described herein. In still further examples, the shot timer **102** may be a standalone computer having computer-executable instructions stored thereon that perform the functions described herein. In additional examples, the shot timer **102** may be more than one device, software module, computer, or device. The present disclosure includes these and other configurations of the shot timer **102**.

[0013] The range **100** includes a firing line **104**. Users **106A**, **106B**, and **106C** (hereinafter referred to individually as “the user **106A**,” “the user **106B**,” and “the user **106C**,” and collectively as “the users **106**”) are typically instructed to stay behind the firing line **104** in area **108** to help maintain the safety of the users **106** while firearms are loaded and active on the range **100**. The users **106** may discharge their firearms (not illustrated) to shoot at targets **110A**, **110B**, and/or **110C** (hereinafter referred to individually as “the target **110A**,” “the target **110B**,” and “the target **110C**,” and collectively as “the targets **110**”). It should be noted that use of the range **100** in FIG. 1 is merely for the purposes of illustrating an example of the present disclosure, as the shot timer **102** may be used in other environments, which are considered to be within the scope of the present disclosure. For example, the shot time **102** may be used in close combat training environments, indoor and outdoor training sessions or environments, and other environments in which the functions of shot detection and timing using the shot timer **102** may be used.

[0014] As noted above, in some examples, if one of the users **106** discharges their firearm at or near the same time as another one of the users **106**, the sound generated by the firearms may arrive at the shot timer **102** at or near the same time. Further, in some examples, there may be other loud noise, such as people yelling and the like, that may be received at the shot timer **102**. Thus, in a first level of shot identification discrimination, the shot timer **102** includes a sound analyzer **112**. The sound analyzer **112** is configured to receive audio signals using a microphone **114** installed on the shot timer **102** or otherwise accessible for use by the shot timer **102**. In some examples, the microphone **114** may be a component of another device that transmits the audio signals to the sound analyzer **112** of the shot timer **102** for audio processing. Upon receipt of the audio signal, the sound analyzer **112**, in a first level of shot detection discrimination, analyzes the audio signal to

determine if the audio signal represents the sound generated by the discharge of a firearm, explained by way of example in FIG. 2.

[0015] FIG. 2 illustrates an audio signal **202** representing audio generating by a firearm and an audio signal **204** generated by a person yelling near the shot timer **102**, in accordance with one or more examples of the presently disclosed subject matter. Both the audio signals **202** and **204** are analyzing using an XY grid, with the X axis representing sound intensity in decibels (dB) and the Y axis representing the sound over a period of time. As illustrated in FIG. 2, while having similar intensities, the audio signal **202** of the firearm being discharged has a different waveform than the audio signal **204** of the person yelling. For example, the audio for a second discharge of a firearm, starting at time **206** reaches a maximum amplitude in a shorter period of time, indicated by time **208**. By contrast, the audio for the beginning of a shout or yell, indicated at time **210**, and the time to reach a maximum amplitude, indicated at time **212**, is longer than the time between time **206** and time **208** of the audio signal **202**. In this example, because the audio signal **204** does not include a rise in amplitude similar to the audio signal **202**, if received by the shot timer **102**, the shot timer **102** may disregard the audio signal **204**.

[0016] It should be noted that there are other technologies for distinguishing between an audio signal generated by a discharged firearm and other audio signals, which are considered to be within the scope of the present disclosure. Returning to FIG. 1, for example, the shot timer **102** may include a sound data store **118**. The sound data store **118** may be a database having stored therein one or more audio signals representing different sounds. These different sounds can be used by the shot timer **102** to match a sound signal received by the microphone **114** to a sound stored in the sound data store. The matching operation may be done for various reasons. For example, a received sound may be matched against a previously stored sound to identify a model of firearm discharged, as different firearms may have different sound signals specific to the particular firearm. Further, the stored sound signals may be used to verify an initial determination that an audio signal received by the shot timer **102** represents the sound generated when the firearm was discharged, described by way of example in FIG. 3.

[0017] FIG. 3 illustrates the audio signal **202** representing audio generating by a firearm and an audio signal **302** stored in the sound data store **118**, in accordance with one or more examples of the presently disclosed subject matter. The audio signal **302** includes audio reference points **304**, **306**, **308**, **310**, and **312**, which are portions of the audio signal **302** at various times, beginning at the beginning at reference point **304** and ending at reference point **312**. The sound analyzer **112** identifies reference points **206**, **208**, **314**, **316**, and **318** on the audio signal **202**, which was received at the microphone **114** of the shot timer **102**. The sound analyzer **112** compares the reference points of the audio signal **202** against the audio signal **302** and can determine that the audio signal **202** was generated by the firearm used to generate the audio signal **302** if the reference points are within a predetermined time and/or intensity. In some examples, the audio signals stored in the sound data store **118** can include audio signals generated from various sources such as the discharging of firearms and other sources such as an explosion, human sounds, and other sound sources. The audio signals stored in the sound data store **118** can be provided by a third-party service that records audio signals from various sources. In other examples, the audio signals stored in the sound data store **118** can be generated and identified by a user of the shot timer **102**, shown by way of example in FIG. 4.

[0018] FIG. 4 is an illustration of a shot sound user interface **402** configured to receive inputs to identify a sound signal for storage in the sound data store **118**, in accordance with one or more examples of the presently disclosed subject matter. The shot sound user interface **402** may be generated and displayed by the shot timer **102** to allow a user to input data relating to a specific sound signal received at the shot timer **102**. In the example illustrated in FIG. 4, a sound signal **404** is provided and displayed in the user interface **402**. The sound signal **404** is audio received at the shot timer **102**. The user interface **402** includes a sound identifier **406** that the user can move to

various parts of the audio signal **404**. In FIG. **4**, the sound identifier **406** has been placed over sound portion **408** of the audio signal **404**. The user interface **402** includes firearm input **410** and ammunition input **412**. The firearm input **410** is configured to receive an input indicating the type of firearm used to generate the sound portion **408**. The ammunition input **412** is configured to receive an input indicating the type of ammunition used by the firearm indicated using the firearm input **410** to generate the sound portion **408**. Further, in some configurations, a user may input their name **414** when using the user interface **402**. Thus, using the data received the shot timer **102** can store the name of the person inputting the information (which may also be the person discharging the firearm), the time in which a firearm was discharged, the type of firearm that generated the audio signal **404**, and the ammunition used when discharging the firearm.

[0019] Returning to FIG. **1**, along with the first layer of discrimination (or verification), e.g., the analysis of sounds by the sound analyzer **112**, the shot timer **102** performs a second layer of verification. In some examples, the shot timer **102** uses pre-discharge visual processing technologies to discriminate between individuals having body movements/positions indicative of a person ready to or currently discharging their firearm and other individuals either at rest or not discharging their firearms. In some examples, a camera of the hybrid shot timer uses skeletal tracking to identify and track those body movements as well as movements of the firearm, explained by way of example in FIG. **5**.

[0020] FIG. **5** illustrates the use of skeleton tracking to determine which, if any, individual discharged a firearm, in accordance with one or more examples of the presently disclosed subject matter. In some examples, skeleton tracking uses sensors, such as a camera **120** illustrated in FIG. **1**. In some examples, the camera **120** may be one or more cameras capable of tracking the motion of a human being. The camera **120** differentiates a human from the background and then identifies features of the human such as joints, shoulders, knees, elbows and hands. Once those joints are identified, in some examples, a posture analyzer **122** uses the features and connects them to form a humanoid skeleton capable of being tracked. It should be noted that other technologies may be used by the shot timer **102** to determine and track the movement of one or more human bodies and are considered to be within the scope of the present disclosure. For example, skeletal tracking software using a machine learning approach can be used. Images or depth images can be labelled to create training data sets where the position of the joints and skeleton has been marked manually, and then from there, that data is used to train the software so that it can recognize joint positions or multiple people in real time.

[0021] FIG. **5** includes a skeleton FIG. **502** and a skeleton FIG. **504**. The skeleton FIG. **502** is formed from features, such as features **506A**, **506B**, **506C**, **506D**, and **506E**, identified for a first user. The skeleton FIG. **504** is formed from features, such as features **508A**, **508B**, **508C**, **508D**, and **508E**, identified for a second user. A posture analyzer **122** illustrated in FIG. **1** is configured to analyze the features and determine if the user is in a standby or non-firing position, or if the user is in a ready or firing position. Users determined to be in a standby or non-firing position can be eliminated as potential users that discharged a firearm. In FIG. **5**, the first user is determined by the posture analyzer **122** as being in a standby or non-firing position, indicated by a firearm being directed towards the sky or upward, shown by the features **506C**, **506D**, and **506E**. By way of contrast, the second user is determined by the posture analyzer **122** as being in ready or firing position, indicated by a firearm being directed towards a target, shown by the features **508C**, **508D**, and **508E**. In this example, upon receiving a sound signal indicating the discharge of a firearm, the shot timer **102** may eliminate the first user as being the user that discharged the firearm. Because the second user is in a ready or firing position, upon receiving the sound signal indicating the discharge of a firearm, the shot timer **102** may indicate that the second user may have discharged their firearm. Thus, in the second level, the first user is eliminated as a potential source of the sound signal. However, in some examples, a user being in a ready or firing position may not be dispositive as being the user that discharged a firearm, as the user may simply be in the position

without discharging the firearm resulting in the sound signal received by the shot timer **102**. Thus, in some examples, a third level using post-discharge firearm or posture behavior may be used, described in FIG. **6**.

[0022] FIG. **6** illustrates a firearm **602** in a pre-discharge position **604** and a post-discharge position **606**, in accordance with one or more examples of the presently disclosed subject matter. The firearm **602** position is illustrated in reference to axis AB, which may be a direction generally from a user (position A) towards a target (position B). A weapon position analyzer **124** illustrated in FIG. **1** may be configured to detect the position of the firearm **602** using the camera **120**. In FIG. **6**, the pre-discharge position **604** indicates that the firearm **602** is in a ready to fire position, whereby the firearm **602** is pointed at the target. Further, in FIG. **6**, the post-discharge position **606** indicates that the firearm **602** has rotated along curve C due to the recoil caused by the discharging of the firearm **602**. Thus, if the shot timer **102** received an audio signal, the shot timer **102** can eliminate firearms that, during the time of the generation of the audio signal, have not moved from the pre-discharge position **604**, indicating that the firearm has not been discharged and therefore, could not be the source of the audio signal. In a similar manner, if the shot timer **102** received an audio signal, the shot timer **102** can include firearms that, during the time of the generation of the audio signal, have moved from the pre-discharge position **604** to the post-discharge position **606**. Other visual cues may be used to determine if the firearm **602** has been discharged, such as, but not limited to, a detection of a slide of the firearm **602** moving, detecting a shell casing being ejected from the firearm **602**, and the like. Another visual cue to determine if the firearm **602** has been discharged may be the detection of a muzzle flash, described further in FIG. **1**.

[0023] Returning to FIG. **1**, the shot timer **102** may further include a flash detector **126**. When some firearms discharge, the firearms can suddenly release high-temperature, high-pressure gases from the muzzle of the firearm. These gases, along with the burning of any remaining gunpowder, can sometimes be seen as intense sources of light, i.e., a muzzle flash. In a manner similar to the movement of the firearm **602** in FIG. **6** moving from the pre-discharge position **604** to the post-discharge position **606**, if available, the flash detector **126** can be used to detect which firearm has been discharged during the period in which an audio signal was generated.

[0024] FIG. **7** illustrates a method **700** for timing discharges of one or more firearms, in accordance with various examples of the presently disclosed subject matter. The method **700** and other processes described herein are illustrated as example flow graphs, each operation of which may represent a sequence of operations that can be implemented in hardware, software, or a combination thereof. In the context of software, the operations represent computer-executable instructions stored on one or more tangible computer-readable storage media that, when executed by one or more processors, perform the recited operations. Generally, computer-executable instructions include routines, programs, objects, components, data structures, and the like that perform particular functions or implement particular abstract data types. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described operations can be combined in any order and/or in parallel to implement the processes.

[0025] The method **700** commences at step **702**, wherein the shot timer **102** is powered on. In some examples, the shot timer **102** may be a stand-alone device having its own power supply. In other examples, the shot timer **102** may be an application executed on a user device, such as a cell phone, and thus is powered on when the application is executed on the user device. In still further examples, the shot timer **102** may be one or more applications running on one or more user devices and in communication with a central service.

[0026] At step **704**, the shot timer **102** is in a standby mode. As used herein, a “standby mode” is a operational condition of the shot timer **102** in which the shot timer **102** is powered on and ready to accept an input.

[0027] At step **706**, the shot timer **102** timing is commenced. The timing may be commenced using various methods. For example, a user may provide an input to cause the shot timer **102** to produce

an audible signal notifying users of the initiation of the shot timer **102**. In other examples, the shot timer **102** may receive an audible input, such as verbal command or sound that indicates an intent to start the internal timer of the shot timer **102**. The present disclosure is not limited to any particular method of commencing the timer of the shot timer **102**.

[0028] At step **708**, an audio signal is received by the shot timer **102**. The audio signal may be sound detected by the microphone **114** of the shot timer **102**. In other examples, the audio signal may be a sound file or data provided to the shot timer **102**. In this example, the audio signal may be data received by the shot timer **102** in which the shot timer **102** calculates the time to discharge a weapon using only the audio signal.

[0029] At step **710**, the shot timer **102** performs the first level verification process. As described above, the first level verification process involves analyzing audio signals received by the shot timer **102** and disregarding audio signals that are not associated with the discharge of a firearm. The present disclosure can use various technologies to perform the first level verification process. An example was described in FIG. **2**, whereby the shot timer **102** analyzed the waveform of the audio signal to determine if the waveform had characteristics indicative of the discharge of a firearm. Another example was described in FIG. **3**, whereby the shot timer **102** matched the audio signal to known audio signals to determine if the received audio signal has a waveform similar to known audio signals. If the first level verification process at step **710** indicates that the received audio signal is not associated with the discharge of a firearm, the method proceeds to step **712**, where the audio signal is disregarded for the purposes of timing a discharge of a firearm.

[0030] If at step **710**, the shot timer **102** determines that the audio signal is associated with the discharge of a firearm, the method **700** continues to step **714**, where a second level verification process is performed. In some examples, the purpose of the second level verification is to identify individuals that could be associated with the discharge of the firearm that produced the audio signal. An example of the second level verification process is described by way of example in FIG. **5**. As described by way of example in FIG. **5**, the shot timer **102** uses a camera or other technology to recognize a posture of one or more individuals that may have discharged the firearm that produced the audio signal. The shot timer **102** creates a skeletal figure of the users. The skeletal figure is analyzed to determine if the individual has a posture that indicates the individual could have discharged the firearm. If the skeletal figure indicates that the individual is in a standby or nonfiring position, the individual is eliminated from consideration as being an individual that discharged the firearm, producing the audio signal.

[0031] At step **716**, the shot timer **102** analyzes the various aspects of the individuals not eliminated from consideration in step **714**. An example is provided in FIG. **6**, whereby the shot timer **102** detects the movement of a firearm from a pre-discharge position to a post-discharge position. In another example, the shot timer **102** may detect a muzzle flash or other movement of a component of the firearm indicating that the firearm has been discharged. The method **700** thereafter ends after the individual that discharged the firearm producing the audio signal is identified. It should be noted that present disclosure does not require the performance of the three levels of verification. For example, in some uses, the different levels of verification may be used alone or in various combinations with other levels of verification.

[0032] FIG. **8** depicts a component level view of the shot timer **102** for use with the systems and methods described herein, in accordance with various examples of the presently disclosed subject matter. The shot timer **102** could be any device capable of providing the functionality associated with the systems and methods described herein. The shot timer **102** can comprise several components to execute the above-mentioned functions. The shot timer **102** may be comprised of hardware, software, or various combinations thereof. The shot timer **102** can be an application operating on a user device, a device configured to perform the functions described herein, a system or network configured to perform the functions described herein, or combinations thereof. As discussed below, the shot timer **102** can comprise memory **802** including an operating system (OS)

804 and one or more standard applications **806**. The standard applications **806** may include applications such as, but not limited to, the sound analyzer **112**, the posture analyzer **122**, the weapon position analyzer **124**, and the flash detector **126**.

[0033] The shot timer **102** can also comprise one or more processors **810** and one or more of removable storage **812**, non-removable storage **814**, transceiver(s) **816**, output device(s) **818**, and input device(s) **820**. In various implementations, the memory **802** can be volatile (such as random access memory (RAM)), non-volatile (such as read only memory (ROM), flash memory, etc.), or some combination of the two. The memory **802** can include data pertaining to an audio signal **850** received by the microphone **114**.

[0034] The memory **802** can also include the OS **804**. The OS **804** varies depending on the manufacturer of the shot timer **102**. The OS **804** contains the modules and software that support basic functions of the shot timer **102**, such as scheduling tasks, executing applications, and controlling peripherals. The OS **804** can also enable the shot timer **102** to send and retrieve other data and perform other functions.

[0035] The shot timer **102** can also comprise one or more processors **810**. In some implementations, the processor(s) **810** can be one or more central processing units (CPUs), graphics processing units (GPUs), both CPU and GPU, or any other combinations and numbers of processing units. The shot timer **102** may also include additional data storage devices (removable and/or non-removable) such as, for example, magnetic disks, optical disks, or tape. Such additional storage is illustrated in FIG. **8** by removable storage **812** and non-removable storage **814**.

[0036] Non-transitory computer-readable media may include volatile and nonvolatile, removable and non-removable tangible, physical media implemented in technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. The memory **802**, removable storage **812**, and non-removable storage **814** are all examples of non-transitory computer-readable media. Non-transitory computer-readable media include, but are not limited to, RAM, ROM, electronically erasable programmable ROM (EEPROM), flash memory or other memory technology, compact disc ROM (CD-ROM), digital versatile discs (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other tangible, physical medium which can be used to store the desired information, which can be accessed by the shot timer **102**. Any such non-transitory computer-readable media may be part of the shot timer **102** or may be a separate database, databank, remote server, or cloud-based server.

[0037] In some implementations, the transceiver(s) **816** include any transceivers known in the art. In some examples, the transceiver(s) **816** can include wireless modem(s) to facilitate wireless connectivity with other components, the Internet, and/or an intranet. Thus, the transceiver(s) **816** can include multiple single-channel transceivers or a multi-frequency, multi-channel transceiver to enable the shot timer **102** to send and receive video calls, audio calls, messaging, etc. The transceiver(s) **816** can enable the shot timer **102** to connect to multiple networks including, but not limited to 2G, 3G, 4G, 5G, and Wi-Fi networks. The transceiver(s) **816** can also include one or more transceivers to enable the shot timer **102** to connect to future (e.g., 6G) networks, Internet-of-Things (IoT), machine-to-machine (M2M), and other current and future networks.

[0038] The transceiver(s) **816** may also include one or more radio transceivers that perform the function of transmitting and receiving radio frequency communications via an antenna (e.g., Wi-Fi or Bluetooth®). In other examples, the transceiver(s) **816** may include wired communication components, such as a wired modem or Ethernet port, for communicating via one or more wired networks. The transceiver(s) **816** can enable the shot timer **102** to facilitate audio and video calls, download files, access web applications, and provide other communications associated with the systems and methods, described above.

[0039] In some implementations, the output device(s) **818** include any output devices known in the art, such as a display (e.g., a liquid crystal or thin-film transistor (TFT) display), a touchscreen,

speakers, a vibrating mechanism, or a tactile feedback mechanism. Thus, the output device(s) can include a screen or display. The output device(s) **818** can also include speakers, or similar devices, to play sounds or ringtones when an audio call or video call is received. Output device(s) **818** can also include ports for one or more peripheral devices, such as headphones, peripheral speakers, or a peripheral display.

[0040] In various implementations, input device(s) **820** include any input devices known in the art. For example, the input device(s) **820** may include a camera, a microphone, or a keyboard/keypad. The input device(s) **820** can include a touch-sensitive display or a keyboard to enable users to enter data and make requests and receive responses via web applications (e.g., in a web browser), make audio and video calls, and use the standard applications **806**, among other things. A touch-sensitive display or keyboard/keypad may be a standard push button alphanumeric multi-key keyboard (such as a conventional QWERTY keyboard), virtual controls on a touchscreen, or one or more other types of keys or buttons, and may also include a joystick, wheel, and/or designated navigation buttons, or the like. A touch sensitive display can act as both an input device **820** and an output device **818**.

[0041] Some aspects of the present disclosure may be described in one or more of the following clauses and in various combinations of the following clauses:

[0042] Clause 1. A method of timing a discharge of a firearm, comprising: initiating a timer of a shot timer; receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of the firearm; and outputting a time based on the first level, the second level, and the third level of analysis.

[0043] Clause 2. The method of clause 1, wherein initiating the timer comprises sounding a signal to indicate to one or more individuals to commence discharging a firearm.

[0044] Clause 3. The method of clause 1, wherein initiating the timer comprises receiving an input that a training session has commenced.

[0045] Clause 4. The method of clause 1, wherein the audio signal is received at a microphone of the shot timer.

[0046] Clause 5. The method of clause 1, wherein performing the first level of analysis by analyzing the audio signal comprises: identifying a starting time of a sound comprising the audio signal; identifying a first time for the sound to reach a maximum amplitude; and analyzing the starting time and the first time to determine that the sound was generated by a firearm.

[0047] Clause 6. The method of clause 1, wherein performing the first level of analysis by analyzing the audio signal comprises matching a sound comprising the audio signal to a sound stored in a sound data store.

[0048] Clause 7. The method of clause 1, wherein performing the second level of analysis by analyzing the posture change of the individual comprises: generating a skeleton figure of the individual; determining, using the skeleton figure, if the individual is in a standby position or a ready position; eliminating the individual as a potential source of the audio signal if the individual is in a standby position.

[0049] Clause 8. The method of clause 1, wherein performing the third level of analysis by analyzing the movement of the firearm comprises determining that the firearm has moved to indicate a recoil of the firearm.

[0050] Clause 9. The method of clause 1, wherein performing the third level of analysis by analyzing the movement of the firearm comprises detecting a muzzle flash of the firearm.

[0051] Clause 10. The method of clause 1, further comprising receiving an input that the audio signal was generated by a type of firearm.

[0052] Clause 11. The method of clause 1, further comprising receiving an input that the audio signal was generated by a type of ammunition.

[0053] Clause 12. The method of clause 1, further comprising receiving an input that the audio

signal was generated by a specific individual.

[0054] Clause 13. A shot timer comprising: a memory storing computer-executable instructions; and a processor in communication with the memory, the computer-executable instructions causing the processor to perform acts comprising: initiating a timer of a shot timer; receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of a firearm; and outputting a time based on the first level, the second level, and the third level of analysis.

[0055] Clause 14. The shot timer of clause 13, wherein the computer-executable instructions causing the processor to perform acts comprising initiating the timer comprises sounding a signal to indicate to one or more individuals to commence discharging a firearm.

[0056] Clause 15. The shot timer of clause 13, wherein the computer-executable instructions causing the processor to perform acts comprising initiating the timer comprises receiving an input that a training session has commenced.

[0057] Clause 16. The shot timer of clause 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the first level of analysis by analyzing the audio signal comprises: identifying a starting time of a sound comprising the audio signal; identifying a first time for the sound to reach a maximum amplitude; and analyzing the starting time and the first time to determine that the sound was generated by a firearm.

[0058] Clause 17. The shot timer of clause 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the first level of analysis by analyzing the audio signal comprises matching a sound comprising the audio signal to a sound stored in a sound data store.

[0059] Clause 18. The shot timer of clause 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the second level of analysis by analyzing the posture change of the individual comprises: generating a skeleton figure of the individual; determining, using the skeleton figure, if the individual is in a standby position or a ready position; and eliminating the individual as a potential source of the audio signal if the individual is in a standby position.

[0060] Clause 19. A non-transitory computer-readable media storing computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of a firearm; and outputting a time based on the first level, the second level, and the third level of analysis.

[0061] Clause 20. The non-transitory computer-readable media of clause 19, further comprising computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving an input that the audio signal was generated by a type of firearm; receiving an input that the audio signal was generated by a type of ammunition; or receiving an input that the audio signal was generated by a specific individual.

[0062] Unless explicitly excluded, the use of the singular to describe a component, structure, or operation does not exclude the use of plural such components, structures, or operations or their equivalents. As used herein, the word “or” refers to any possible permutation of a set of items. For example, the phrase “A, B, or C” refers to at least one of A, B, C, or any combination thereof, such as any of: A; B; C; A and B; A and C; B and C; A, B, and C; or multiple of any item such as A and A; B, B, and C; A, A, B, C, and C; etc.

[0063] While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed systems and methods without departing from the spirit and scope of what is disclosed. Such embodiments

should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

- 1.** A method of timing a discharge of a firearm, comprising: initiating a timer of a shot timer; receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of the firearm; and outputting a time based on the first level, the second level, and the third level of analysis.
- 2.** The method of claim 1, wherein initiating the timer comprises sounding a signal to indicate to one or more individuals to commence discharging a firearm.
- 3.** The method of claim 1, wherein initiating the timer comprises receiving an input that a training session has commenced.
- 4.** The method of claim 1, wherein the audio signal is received at a microphone of the shot timer.
- 5.** The method of claim 1, wherein performing the first level of analysis by analyzing the audio signal comprises: identifying a starting time of a sound comprising the audio signal; identifying a first time for the sound to reach a maximum amplitude; and analyzing the starting time and the first time to determine that the sound was generated by a firearm.
- 6.** The method of claim 1, wherein performing the first level of analysis by analyzing the audio signal comprises matching a sound comprising the audio signal to a sound stored in a sound data store.
- 7.** The method of claim 1, wherein performing the second level of analysis by analyzing the posture change of the individual comprises: generating a skeleton figure of the individual; determining, using the skeleton figure, if the individual is in a standby position or a ready position; eliminating the individual as a potential source of the audio signal if the individual is in a standby position.
- 8.** The method of claim 1, wherein performing the third level of analysis by analyzing the movement of the firearm comprises determining that the firearm has moved to indicate a recoil of the firearm.
- 9.** The method of claim 1, wherein performing the third level of analysis by analyzing the movement of the firearm comprises detecting a muzzle flash of the firearm.
- 10.** The method of claim 1, further comprising receiving an input that the audio signal was generated by a type of firearm.
- 11.** The method of claim 1, further comprising receiving an input that the audio signal was generated by a type of ammunition.
- 12.** The method of claim 1, further comprising receiving an input that the audio signal was generated by a specific individual.
- 13.** A shot timer comprising: a memory storing computer-executable instructions; and a processor in communication with the memory, the computer-executable instructions causing the processor to perform acts comprising: initiating a timer of a shot timer; receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of a firearm; and outputting a time based on the first level, the second level, and the third level of analysis.
- 14.** The shot timer of claim 13, wherein the computer-executable instructions causing the processor to perform acts comprising initiating the timer comprises sounding a signal to indicate to one or more individuals to commence discharging a firearm.
- 15.** The shot timer of claim 13, wherein the computer-executable instructions causing the processor to perform acts comprising initiating the timer comprises receiving an input that a training session has commenced.

- 16.** The shot timer of claim 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the first level of analysis by analyzing the audio signal comprises: identifying a starting time of a sound comprising the audio signal; identifying a first time for the sound to reach a maximum amplitude; and analyzing the starting time and the first time to determine that the sound was generated by a firearm.
- 17.** The shot timer of claim 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the first level of analysis by analyzing the audio signal comprises matching a sound comprising the audio signal to a sound stored in a sound data store.
- 18.** The shot timer of claim 13, wherein the computer-executable instructions causing the processor to perform acts comprising performing the second level of analysis by analyzing the posture change of the individual comprises: generating a skeleton figure of the individual; determining, using the skeleton figure, if the individual is in a standby position or a ready position; and eliminating the individual as a potential source of the audio signal if the individual is in a standby position.
- 19.** A non-transitory computer-readable media storing computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving an audio signal; performing a first level of analysis by analyzing the audio signal; performing a second level of analysis by analyzing a posture change of an individual; performing a third level of analysis by analyzing a movement of a firearm; and outputting a time based on the first level, the second level, and the third level of analysis.
- 20.** The non-transitory computer-readable media of claim 19, further comprising computer-executable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving an input that the audio signal was generated by a type of firearm; receiving an input that the audio signal was generated by a type of ammunition; or receiving an input that the audio signal was generated by a specific individual.
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